Cloud Computing for Agri-Food

Computing and Mathematics

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Cloud Computing for Agri-Food (A21781)

Short Title:	Cloud Computing for Agri-Food		
Faculty	Science and Computing		
Department	Computing and Mathematics		
Cluster	AgriFoodICT		
Author	ADUNPHY		
Credits	10	**Level:**	Advanced

Description of Module / Aims

This module introduces students to the capabilities of cloud computing, especially when applied to agricultural applications. The main concepts of cloud computing are covered and students carry out a series of practical exercises with cloud computing technologies and services. Students are also exposed to those concepts of computer organization, security and networking that are required to work in cloud environments.

Programmes

`COMP-0980`	HDip in Science in Agri-Food ICT Systems (`WD_SAFICT_G`)	5 / 2 / M
	Higher Diploma in Science in Agi-Food ICT Systems (`WD_18AFICT_G`)	1 / 2 / M

Indicative Content

- Main concepts of cloud computing
- Overview of computer systems infrastructure technologies: number bases; computer organisation; operating systems; file management
- Computer networks: protocols & encapsulation; addressing and identification; LANs & WANs, TCP/IP, HTTP, DNS
- Security concepts -- encryption, authentication, public key infrastructure.
- Virtualisation
- Cloud usage scenarios
- Using and integrating cloud services
- Public cloud services: storage; compute; networking
- Configuration of multi-tier application infrastructure services
- Virtual private clouds
- Web Application Architectures -- Performance, Scaling, Load Balancing and Security
- Network and application management and monitoring

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Discuss the evolution of cloud computing.
2. Appraise the main computing service and deployment models that are in widespread use.
3. Install and set up a contemporary operating system (e.g. within a virtualised or single board environment), and configure the services necessary to support basic applications.
4. Analyse the operation and features of computer network protocols and services.
5. Demonstrate competency with a limited set of the utilities provided by a contemporary operating system using a command-line interface (CLI).
6. Deploy an application that integrates a selection of cloud services using a popular cloud platform (e.g. IBM Bluemix or Amazon Web Services).
7. Explore cloud computing services and technologies from the perspectives of security, privacy and cost.
8. Compare and contrast the main technologies required to manage and monitor cloud-based applications.
9. Appraise and apply principles of application performance, scalability, load balancing and security.

Learning and Teaching Methods

- This module will be presented by a combination of lectures and practicals.
- The lectures will be used to introduce new topics and their related concepts.
- The practical lab component will be delivered in block sessions.
- Strong emphasis is given to practical laboratory exercises with extensive use made of virtualised and cloud environments.

Assessment Methods

	Weighing		Outcomes Assessed
Final Written Examination	50%		1,2,4,8,9
Continuous Assessment	50%		
Practical		50%	3,5,6,7,9

Assessment Criteria

<40%: Unable to explain key cloud computing concepts, technologies and services. Unable to effectively use a cloud platform. Unable to configure basic infrastructure services to meet assignment requirements. Unable to interpret and describe key concepts of cloud application infrastructure and automation.

40%-49%: Able to explain key cloud computing concepts, technologies and services and carry out basic analysis. Can deploy an application to a popular cloud platform. Can configure basic infrastructure services to meet assignment requirements. Able to interpret and describe key concepts of cloud application services and automation.

50%-59%: In addition, can explain in technical detail key cloud computing technologies and services. Can deploy a moderately sophisticated application to a popular cloud platform. Can discuss key concepts of the specific knowledge domains covered above and ability to discover and integrate related knowledge into cloud based application architectures.

60%-69%: In addition, can deploy a complex application to a popular cloud platform. Able to solve problems within the specific knowledge domain by experimenting with the appropriate skills and tools.

70%-100%: All of the above to an excellent level. Can go beyond the basic practical specification and demonstrate competence with other services. Can demonstrate a deep understanding of deployment and management of a multi-tier cloud application.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	60	
Independent Learning	174	

Essential Material

- "Amazon Web Services." "Amazon Web Services." https://aws.amazon.com/

Supplementary Material

- "IBM Bluemix." https://www.ibm.com/cloud-computing/bluemix/
- Amazon, A. _Getting started with AWS_ . New York:Amazon : ebook, 2014.
- Morris, K. _Infrastructure as Code: Managing Servers in the Cloud_. New York: O'Rielly Media, 2016.
- Rafaels, R. _Cloud Compu__tiog: From Beginning to End_. 1st. New York:CreateSpace: Independant Publishing Platform, 2015.

Requested Resources

- COMPUTER LAB: BYOD Lab